MATHEMATICS / INDEPENDENT STUDY

MM5020

Mathematics III - Abstract Algebra

Progressive hints

Original teaching material aligned with Stockholm University course descriptions.
Independently authored; not an official university publication.

Edition 1 · 24 October 2026

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How to use this volume E problems practice definitions and standard arguments; C problems combine ideas; A problems require synthesis or generalization. These are study difficulty labels, not official SU grading criteria. Time estimates are guides, not deadlines. Write a complete attempt before opening hints. In the hints volume, read Hint 1 first and return to the problem before consulting Hint 2. Solutions give a decisive idea, a complete argument, and criteria for self-assessment.

1 Unit 1 · Groups and cyclic structure

MM5020-U1.01 · E · 15 min
Learning objective: U1.1 Group axioms

Show that the even integers form a group under addition, and explain why they do not form a group under multiplication.

Hint 1 — a starting point. List the identity and inverse for addition.

Hint 2 — the key move. The multiplicative identity would have to be one.

MM5020-U1.02 · E · 15 min
Learning objective: U1.1 Cancellation

Solve $axb = c$ in an arbitrary group and prove uniqueness.

Hint 1 — a starting point. Undo the left factor on the left.

Hint 2 — the key move. Multiply first by a^{-1} on the left and then by b^{-1} on the right.

MM5020-U1.03 · E · 15 min
Learning objective: U1.2 Element orders

Find the order and generated subgroup of $[9]$ in $\mathbb{Z}/15\mathbb{Z}$.

Hint 1 — a starting point. List successive multiples or use the gcd formula.

Hint 2 — the key move. The order is $\frac{15}{\gcd(15,9)}$.

MM5020-U1.04 · E · 15 min
Learning objective: U1.2 Subgroup test

Prove that the intersection of two subgroups of a group is a subgroup.

Hint 1 — a starting point. The intersection contains the identity.

Hint 2 — the key move. Apply the one-step subgroup test in both subgroups.

MM5020-U1.05 · C · 30 min
Learning objective: U1.1 Structural identities

Prove that a group in which $x^2 = e$ for every x is abelian.

Hint 1 — a starting point. Every element is its own inverse.

Hint 2 — the key move. Compare $(ab)^{-1}$ with $b^{-1}a^{-1}$.

MM5020-U1.06 · C · 30 min
Learning objective: U1.2 Cyclic subgroups

List all subgroups of $\mathbb{Z}/18\mathbb{Z}$ and their orders.

Hint 1 — a starting point. Subgroups of a cyclic group correspond to divisors of its order.

Hint 2 — the key move. For each divisor d of eighteen, use $\langle [d] \rangle$.

MM5020-U1.07 · C · 30 min
Learning objective: U1.2 Generator arithmetic
Let a have order n. Prove $\langle a^r \rangle = \langle a^{\gcd(n,r)} \rangle$.
Hint 1 — a starting point. Use a Bézout identity for n, r.
Hint 2 — the key move. If $d = un + vr$, then $a^d = (a^r)^v$.

MM5020-U1.08 · C · 30 min
Learning objective: U1.2 Subgroup structure
Prove that the union of two subgroups H, K is a subgroup exactly when one contains the other.
Hint 1 — a starting point. Assume neither contains the other and choose one element from each difference.
Hint 2 — the key move. For $h \in H \setminus K$ and $k \in K \setminus H$, test hk.

MM5020-U1.09 · C · 30 min
Learning objective: U1.1 Noncommutative examples
Let r rotate a square by a quarter turn and s reflect it in a vertical axis. Show $srs = r^{-1}$ and use this to prove the symmetry group is nonabelian.
Hint 1 — a starting point. Follow a vertex through the three motions.
Hint 2 — the key move. Reflection reverses orientation, hence reverses the direction of rotation.

MM5020-U1.10 · A · 50 min
Learning objective: U1.2 Automorphisms
Classify all automorphisms of the additive group $\mathbb{Z}/n\mathbb{Z}$.
Hint 1 — a starting point. A homomorphism is determined by the image of $[1]$.
Hint 2 — the key move. The image must generate the group.

MM5020-U1.11 · A · 50 min
Learning objective: U1.2 Products of cyclic groups
Prove that $C_m \times C_n$ is cyclic exactly when $\gcd(m, n) = 1$.
Hint 1 — a starting point. Find the order of a pair of elements.
Hint 2 — the key move. A pair of generators has order $\operatorname{lcm}(m, n)$.

MM5020-U1.12 · A · 50 min
Learning objective: U1.1 Axioms and finiteness
Let M be a finite nonempty set with an associative operation and a two-sided identity. Assume left and right cancellation. Prove that M is a group.
Hint 1 — a starting point. Multiplication by a fixed element is injective.
Hint 2 — the key move. On a finite set it is surjective, so left and right inverses exist.

2 Unit 2 · Cosets and permutations

MM5020-U2.01 · E · 15 min
Learning objective: U2.1 Cosets
List the cosets of $H = \{[0], [4], [8]\}$ in $\mathbb{Z}/12\mathbb{Z}$.
Hint 1 — a starting point. Translate H by small residues.
Hint 2 — the key move. There are $\frac{12}{3} = 4$ cosets.

MM5020-U2.02 · E · 15 min
Learning objective: U2.1 Lagrange
What element orders are possible in a group of order twenty-one according to Lagrange's theorem? Does this alone guarantee an element of order twenty-one?
Hint 1 — a starting point. List the positive divisors.
Hint 2 — the key move. A necessary condition need not be sufficient.

MM5020-U2.03 · E · 15 min
Learning objective: U2.2 Permutation composition
Compute $(123)(23)$ and its sign, composing right to left.
Hint 1 — a starting point. Track the images of one, two, and three.
Hint 2 — the key move. One maps to two, two maps to one, and three is fixed.

MM5020-U2.04 · E · 15 min
Learning objective: U2.2 Cycle order
Find the order and sign of $(1234)(567)$ in S_7.
Hint 1 — a starting point. The cycles are disjoint.
Hint 2 — the key move. Take the least common multiple and multiply their signs.

MM5020-U2.05 · C · 30 min
Learning objective: U2.1 Coset criteria
Prove $aH = bH$ if and only if $a^{-1}b \in H$.
Hint 1 — a starting point. Start with membership of b in aH.
Hint 2 — the key move. Write $b = ah$.

MM5020-U2.06 · C · 30 min
Learning objective: U2.1 Prime order
Prove that a group of prime order has no proper nontrivial subgroup and is cyclic.
Hint 1 — a starting point. Apply Lagrange twice.
Hint 2 — the key move. The subgroup generated by a nonidentity element has order greater than one.

MM5020-U2.07 · C · 30 min
Learning objective: U2.2 Powers of cycles
Describe the disjoint cycles of $(123456)^2$ and $(123456)^3$, and compute their orders.
Hint 1 — a starting point. Adding the exponent modulo six describes the action on positions.
Hint 2 — the key move. The cycles are the orbits of adding two or three.

MM5020-U2.08 · C · 30 min
Learning objective: U2.2 Counting permutations
How many three-cycles are in S_5? How many products of two disjoint transpositions?
Hint 1 — a starting point. Choose supports before choosing arrangements.
Hint 2 — the key move. A three-element support has two cycles; a four-element support has three pairings.

MM5020-U2.09 · C · 30 min
Learning objective: U2.2 Alternating groups
Show every three-cycle is even and every even permutation is a product of three-cycles.
Hint 1 — a starting point. Pair the transpositions in an even factorization.
Hint 2 — the key move. A pair sharing a letter is a three-cycle; a disjoint pair can be split using an intermediate transposition.

MM5020-U2.10 · A · 50 min
Learning objective: U2.2 Minimal permutation degree
Find the least n for which S_n has an element of order fifteen.
Hint 1 — a starting point. Cycle lengths must have least common multiple fifteen.
Hint 2 — the key move. Compare one fifteen-cycle with disjoint cycles of lengths three and five.

MM5020-U2.11 · A · 50 min
Learning objective: U2.1 Indices in a tower
For finite G and $K \leq H \leq G$, prove $[G : K] = [G : H][H : K]$ by constructing coset representatives.
Hint 1 — a starting point. Choose representatives for both stages.
Hint 2 — the key move. Products of representatives give representatives for G/K.

MM5020-U2.12 · A · 50 min
Learning objective: U2.2 Conjugation and cycles
Prove $\tau(a_1 \ldots a_r)\tau^{-1} = (\tau(a_1) \ldots \tau(a_r))$. Deduce that conjugate permutations have the same cycle lengths, and conversely.
Hint 1 — a starting point. Evaluate the conjugate on $\tau(a_i)$.
Hint 2 — the key move. For the converse, choose a bijection matching corresponding cycle positions.

3 Unit 3 · Homomorphisms and quotients

MM5020-U3.01 · E · 15 min
Learning objective: U3.1 Kernels
Find the kernel and image of $f : \mathbb{Z} \to \mathbb{Z}/8\mathbb{Z}$, $f(n) = [3n]$.
Hint 1 — a starting point. Three is invertible modulo eight.
Hint 2 — the key move. Its inverse is also three.

MM5020-U3.02 · E · 15 min
Learning objective: U3.1 Normality
Show the kernel of the determinant map $\mathrm{GL}_2(\mathbb{R}) \to \mathbb{R}^\times$ is normal.
Hint 1 — a starting point. Use the homomorphism property of determinants.
Hint 2 — the key move. The kernel consists of matrices of determinant one.

MM5020-U3.03 · E · 15 min
Learning objective: U3.2 Quotient size
Let $|G| = 60$ and $N \lhd G$ have order twelve. Determine $|G/N|$ and its group structure.
Hint 1 — a starting point. Count cosets.
Hint 2 — the key move. A group of order five is cyclic.

MM5020-U3.04 · E · 15 min
Learning objective: U3.1 Homomorphisms
Show a homomorphism $f : G \to H$ satisfies $f(g^n) = f(g)^n$ for every integer n.
Hint 1 — a starting point. Handle positive, zero, and negative exponents separately.
Hint 2 — the key move. Use preservation of inverses for negative exponents.

MM5020-U3.05 · C · 30 min
Learning objective: U3.2 First isomorphism theorem
Identify $\mathbb{R}^\times/\{1, -1\}$ using a natural homomorphism.
Hint 1 — a starting point. Take the absolute value.
Hint 2 — the key move. Its image is the multiplicative group of positive real numbers.

MM5020-U3.06 · C · 30 min
Learning objective: U3.1 Normality
Prove the center $Z(G)$ is a normal subgroup.
Hint 1 — a starting point. Check the subgroup test, then conjugation.
Hint 2 — the key move. Central elements are fixed by every conjugation.

MM5020-U3.07 · C · 30 min
Learning objective: U3.2 Quotient maps

For positive integers m, n, when is $[a]_m \mapsto [a]_n$ well defined? When it is, find its kernel.
Hint 1 — a starting point. The representatives zero and m must agree after mapping.
Hint 2 — the key move. The necessary and sufficient condition is $n|m$.

MM5020-U3.08 · C · 30 min
Learning objective: U3.2 Correspondence

List all subgroups of $\mathbb{Z}/12\mathbb{Z}$ by using subgroups of $\mathbb{Z}$ containing $12\mathbb{Z}$.
Hint 1 — a starting point. All subgroups of $\mathbb{Z}$ are $d\mathbb{Z}$.
Hint 2 — the key move. Containment $12\mathbb{Z} \subseteq d\mathbb{Z}$ means $d|12$.

MM5020-U3.09 · C · 30 min
Learning objective: U3.1 Cyclic quotient by center

Prove that if $G/Z(G)$ is cyclic, then G is abelian.
Hint 1 — a starting point. Express every element using a representative of the quotient generator.
Hint 2 — the key move. Write $x = g^r z$ and $y = g^s w$ with z, w central.

MM5020-U3.10 · A · 50 min
Learning objective: U3.2 Universal property

Prove that a homomorphism $f : G \to H$ factors through G/N exactly when $N \subseteq \ker f$, and prove uniqueness of the factor.
Hint 1 — a starting point. A factor must kill the kernel of the projection.
Hint 2 — the key move. For sufficiency define $\overline{f}(gN) = f(g)$ and check representatives.

MM5020-U3.11 · A · 50 min
Learning objective: U3.2 Abelianization

Let $[G, G]$ be generated by all commutators $aba^{-1}b^{-1}$. Show it is normal, its quotient is abelian, and every map from G to an abelian group factors uniquely through this quotient.
Hint 1 — a starting point. Conjugation takes commutators to commutators.
Hint 2 — the key move. An abelian target kills each commutator.

MM5020-U3.12 · A · 50 min
Learning objective: U3.1 Failure of converse to Lagrange

Prove that A_4 has no subgroup of order six. You may use that three-cycles generate A_4.
Hint 1 — a starting point. A subgroup of order six would have index two.
Hint 2 — the key move. The quotient would have order two, but every three-cycle must map to its identity.

4 Unit 4 · Actions and simple groups

MM5020-U4.01 · E · 15 min
Learning objective: U4.1 Actions
Verify that S_n acts on $\{1, \dots, n\}$ by evaluation.
Hint 1 — a starting point. Check identity and composition.
Hint 2 — the key move. The operation in S_n is function composition.

MM5020-U4.02 · E · 15 min
Learning objective: U4.1 Stabilizers
Find the stabilizer of one in the natural S_4 action and its order.
Hint 1 — a starting point. It permutes the other three letters freely.
Hint 2 — the key move. It is isomorphic to S_3.

MM5020-U4.03 · E · 15 min
Learning objective: U4.2 Conjugacy
Compute $(123)(12)(123)^{-1}$.
Hint 1 — a starting point. Conjugation relabels a cycle.
Hint 2 — the key move. Send one to two and two to three.

MM5020-U4.04 · E · 15 min
Learning objective: U4.2 Simplicity
Which nontrivial finite cyclic groups are simple?
Hint 1 — a starting point. Every subgroup of an abelian group is normal.
Hint 2 — the key move. A cyclic group of composite order has a proper nontrivial subgroup.

MM5020-U4.05 · C · 30 min
Learning objective: U4.1 Kernel of an action
Show the kernel of the homomorphism associated with an action is the intersection of all stabilizers.
Hint 1 — a starting point. Both descriptions say an element fixes every point.
Hint 2 — the key move. Write the two membership conditions explicitly.

MM5020-U4.06 · C · 30 min
Learning objective: U4.1 Conjugate stabilizers
Prove $G_{h \cdot x} = hG_x h^{-1}$.
Hint 1 — a starting point. Apply h^{-1} to the equation defining a stabilizer.
Hint 2 — the key move. The condition is equivalent to $h^{-1}gh$ fixing x.

MM5020-U4.07 · C · 30 min
Learning objective: U4.2 Class equation
Use conjugacy classes to show the center of S_3 is trivial.
Hint 1 — a starting point. List its cycle types.
Hint 2 — the key move. The class sizes are one, three, and two.

MM5020-U4.08 · C · 30 min
Learning objective: U4.2 Prime-power groups
Prove a group of order p^2 is abelian.
Hint 1 — a starting point. The center has order divisible by p.
Hint 2 — the key move. If its order were p, the quotient by the center would be cyclic.

MM5020-U4.09 · C · 30 min
Learning objective: U4.1 Orbit counting
Count two-colorings of four labeled square vertices up to rotation, with reflections excluded.
Hint 1 — a starting point. Use the cyclic rotation group of order four.
Hint 2 — the key move. The four rotations fix $16, 2, 4, 2$ colorings respectively.

MM5020-U4.10 · A · 50 min
Learning objective: U4.2 Simple prime-power groups
Classify finite simple groups whose order is a power of a prime.
Hint 1 — a starting point. Their center is a nontrivial normal subgroup.
Hint 2 — the key move. Simplicity forces the center to be the whole group.

MM5020-U4.11 · A · 50 min
Learning objective: U4.1 Coset actions
Let H have finite index n in G. Show there is a normal subgroup $N \subseteq H$ with $[G : N] \leq n!$.
Hint 1 — a starting point. Let G act on its left cosets of H.
Hint 2 — the key move. Use the kernel of the resulting map to S_n.

MM5020-U4.12 · A · 50 min
Learning objective: U4.2 Faithful actions of simple groups
Prove every nontrivial action of a simple group is faithful. Deduce that A_5 cannot act nontrivially on a set of four elements.
Hint 1 — a starting point. The kernel of an action is normal.
Hint 2 — the key move. A faithful action on four points would embed A_5 into S_4.

5 Unit 5 · Rings and their quotients

MM5020-U5.01 · E · 15 min
Learning objective: U5.1 Units
Find every unit in $\mathbb{Z}/10\mathbb{Z}$ and its inverse.
Hint 1 — a starting point. A residue is a unit exactly when it is coprime to ten.
Hint 2 — the key move. Test the residues one, three, seven, and nine.

MM5020-U5.02 · E · 15 min
Learning objective: U5.1 Zero divisors
Give nonzero matrices $A, B \in M_2(\mathbb{Q})$ with $AB = 0$.
Hint 1 — a starting point. Use projections onto different coordinate axes.
Hint 2 — the key move. Take diagonal matrices with complementary single nonzero entries.

MM5020-U5.03 · E · 15 min
Learning objective: U5.2 Ideals
Show $6\mathbb{Z}$ is an ideal of $\mathbb{Z}$, and describe the identity of its quotient.
Hint 1 — a starting point. Check additive subgroup and absorption.
Hint 2 — the key move. Multiplying a multiple of six by any integer preserves that form.

MM5020-U5.04 · E · 15 min
Learning objective: U5.1 Quaternions
Find the inverse of $q = 1 + 2i - j$.
Hint 1 — a starting point. Use quaternion conjugation.
Hint 2 — the key move. Its squared norm is six.

MM5020-U5.05 · C · 30 min
Learning objective: U5.2 Quotient criteria
Determine whether (5) and (6) are prime or maximal in $\mathbb{Z}$.
Hint 1 — a starting point. Inspect the quotient rings.
Hint 2 — the key move. A prime-order residue ring is a field; modulo six there are zero divisors.

MM5020-U5.06 · C · 30 min
Learning objective: U5.2 Evaluation
Show $\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}$.
Hint 1 — a starting point. Evaluate a polynomial at i.
Hint 2 — the key move. Divide by $x^2 + 1$ to identify the kernel.

MM5020-U5.07 · C · 30 min
Learning objective: U5.1 Characteristic
Prove a nonzero domain has characteristic zero or a prime.
Hint 1 — a starting point. A composite characteristic factors.
Hint 2 — the key move. Translate the integer factorization into a zero product in the ring.

MM5020-U5.08 · C · 30 min
Learning objective: U5.2 Ideals and units
Prove an ideal contains a unit if and only if it is the whole ring. Deduce that a field has exactly two ideals.
Hint 1 — a starting point. Multiply the unit by its inverse.
Hint 2 — the key move. An ideal containing one contains every element.

MM5020-U5.09 · C · 30 min
Learning objective: U5.2 Noncommutative ideals
Show the matrices with zero second column form a left ideal of $M_2(K)$ but not a two-sided ideal.
Hint 1 — a starting point. Left multiplication preserves a zero column.
Hint 2 — the key move. Multiply E_{11} on the right by E_{12}.

MM5020-U5.10 · A · 50 min
Learning objective: U5.2 Chinese remainder theorem
Prove $\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ for coprime positive m, n.
Hint 1 — a starting point. Use simultaneous reduction and a Bézout identity.
Hint 2 — the key move. The kernel is $mn\mathbb{Z}$; use Bézout to construct any prescribed pair of residues.

MM5020-U5.11 · A · 50 min
Learning objective: U5.2 Matrix ring ideals
Prove every two-sided ideal of $M_{n(K)}$, for a field K, is zero or the whole ring.
Hint 1 — a starting point. Start with a nonzero entry of a matrix in the ideal.
Hint 2 — the key move. Multiplication by matrix units isolates that entry in any desired position.

MM5020-U5.12 · A · 50 min
Learning objective: U5.1 Division rings
Prove quaternion conjugation reverses products and deduce multiplicativity of $N(q) = q\overline{q}$.
Hint 1 — a starting point. It is enough to verify the reversal on the basis elements and extend bilinearly.
Hint 2 — the key move. Use $\overline{qr} = \overline{rq}$ and centrality of real norms.

6 Unit 6 · Polynomials and factorization

MM5020-U6.01 · E · 15 min
Learning objective: U6.1 Division
Divide $x^3 + 2x + 1$ by $x^2 + 1$ over $\mathbb{Q}$.
Hint 1 — a starting point. Cancel the leading term first.
Hint 2 — the key move. Subtract $x(x^2 + 1)$.

MM5020-U6.02 · E · 15 min
Learning objective: U6.1 Bézout
Find integers u, v with $30u + 21v = 3$.
Hint 1 — a starting point. Run the Euclidean algorithm.
Hint 2 — the key move. Use $3 = 21 - 2(30 - 21)$.

MM5020-U6.03 · E · 15 min
Learning objective: U6.2 Irreducibility
Prove $x^3 - 5$ is irreducible over $\mathbb{Q}$.
Hint 1 — a starting point. Use a prime dividing the constant term.
Hint 2 — the key move. Eisenstein applies at five.

MM5020-U6.04 · E · 15 min
Learning objective: U6.1 Roots
Factor $x^2 + 1$ over $\mathbb{F}_5$.
Hint 1 — a starting point. Find a square equal to negative one.
Hint 2 — the key move. Two squared equals four modulo five.

MM5020-U6.05 · C · 30 min
Learning objective: U6.1 Modular inverses
Find the inverse of $x + 1$ in $\mathbb{Q}[x]/(x^2 + 1)$.
Hint 1 — a starting point. Multiply by $x - 1$.
Hint 2 — the key move. The product is $x^2 - 1$, which becomes negative two.

MM5020-U6.06 · C · 30 min
Learning objective: U6.2 Reduction test
Prove $x^3 + x + 1$ is irreducible over $\mathbb{Q}$ by reducing modulo two.
Hint 1 — a starting point. A cubic over a field is reducible exactly when it has a root.
Hint 2 — the key move. Evaluate at the only two elements of $\mathbb{F}_2$.

MM5020-U6.07 · C · 30 min
Learning objective: U6.2 Failure outside PIDs
Show 2 is irreducible but not prime in $\mathbb{Z}\left[\sqrt{-5}\right]$.
Hint 1 — a starting point. Use the norm $N\left(a+b\sqrt{-5}\right)=a^2+5b^2$.
Hint 2 — the key move. There is no element of norm two, but $\left(1+\sqrt{-5}\right)\left(1-\sqrt{-5}\right)=6$.

MM5020-U6.08 · C · 30 min
Learning objective: U6.1 Principal ideals
Show $K[x]/(f)$ is a field if and only if f is irreducible, assuming f is nonconstant.
Hint 1 — a starting point. Use maximal ideals or Bézout.
Hint 2 — the key move. For irreducible f, every polynomial not divisible by f is coprime to it.

MM5020-U6.09 · C · 30 min
Learning objective: U6.2 Primitive products
Show the product of two primitive integer polynomials is primitive.
Hint 1 — a starting point. Suppose a prime divides every coefficient of the product.
Hint 2 — the key move. Reduce modulo that prime and use that $\mathbb{F}_{p[x]}$ is a domain.

MM5020-U6.10 · A · 50 min
Learning objective: U6.2 Shifted Eisenstein
For prime p, prove $1+x+\ldots+x^{p-1}$ is irreducible over $\mathbb{Q}$.
Hint 1 — a starting point. Substitute $x=y+1$.
Hint 2 — the key move. The transformed polynomial is $\frac{(y+1)^p-1}{y}$.

MM5020-U6.11 · A · 50 min
Learning objective: U6.1 Polynomial Chinese remainder
If $f,g\in K[x]$ are coprime, prove $K[x]/(fg)\cong K[x]/(f)\times K[x]/(g)$.
Hint 1 — a starting point. Use simultaneous reduction and polynomial Bézout.
Hint 2 — the key move. Construct a preimage of $([a],[b])$ using $uf+vg=1$.

MM5020-U6.12 · A · 50 min
Learning objective: U6.2 Nonprincipal ideals
Prove $\mathbb{Z}[x]$ is a UFD but not a PID.
Hint 1 — a starting point. Apply Gauss's theorem, then study $(2,x)$.
Hint 2 — the key move. A generator of $(2,x)$ would divide both two and x.

7 Unit 7 · Fields as vector spaces

MM5020-U7.01 · E · 15 min
Learning objective: U7.1 Fractions

Prove $\frac{a}{b} = 0$ in the fraction field of a domain exactly when $a = 0$.
Hint 1 — a starting point. Compare with the fraction $\frac{0}{1}$.
Hint 2 — the key move. Use the cross-multiplication definition.

MM5020-U7.02 · E · 15 min
Learning objective: U7.2 Minimal polynomials

Find the minimal polynomial of $\sqrt{7}$ over $\mathbb{Q}$ and a basis of its generated field.
Hint 1 — a starting point. Use the polynomial $x^2 - 7$.
Hint 2 — the key move. It is irreducible by Eisenstein at seven.

MM5020-U7.03 · E · 15 min
Learning objective: U7.1 Tower law

If $[L : K] = 3$ and $[M : L] = 4$, find $[M : K]$. Can an intermediate field between K and M have degree five over K?
Hint 1 — a starting point. Multiply the degrees.
Hint 2 — the key move. An intermediate degree must divide the total degree.

MM5020-U7.04 · E · 15 min
Learning objective: U7.2 Power-basis arithmetic

With $\alpha^3 = 2$, simplify $\alpha^5 + \alpha^4$ in the power basis.
Hint 1 — a starting point. Reduce each exponent using $\alpha^3 = 2$.
Hint 2 — the key move. The remaining powers should be below three.

MM5020-U7.05 · C · 30 min
Learning objective: U7.2 Inverses

Find $\left(2 + \sqrt{3}\right)^{-1}$ in $\mathbb{Q}\left(\sqrt{3}\right)$ and explain why every nonzero $a + b\sqrt{3}$ has an inverse in this field.
Hint 1 — a starting point. Multiply by the conjugate expression.
Hint 2 — the key move. The denominator is $a^2 - 3b^2$.

MM5020-U7.06 · C · 30 min
Learning objective: U7.2 Algebraicity

Prove every element of a finite extension is algebraic.
Hint 1 — a starting point. Use too many powers in a finite-dimensional vector space.
Hint 2 — the key move. If the degree is d, the $d + 1$ elements $1, \alpha, \ldots, \alpha^d$ are dependent.

MM5020-U7.07 · C · 30 min
Learning objective: U7.2 Change of generator
Find the minimal polynomial of $1+\sqrt{2}$ over $\mathbb{Q}$.
Hint 1 — a starting point. Isolate the square root and square.
Hint 2 — the key move. A quadratic with no rational root is irreducible.

MM5020-U7.08 · C · 30 min
Learning objective: U7.1 Fraction universal property
Explain why the reduction map $\mathbb{Z} \to \mathbb{F}_5$ cannot extend to a unital homomorphism $\mathbb{Q} \to \mathbb{F}_5$.
Hint 1 — a starting point. Consider the inverse of five in $\mathbb{Q}$.
Hint 2 — the key move. A homomorphism would send $5 \cdot \left(\frac{1}{5}\right) = 1$ to zero.

MM5020-U7.09 · C · 30 min
Learning objective: U7.2 Root adjunction
Construct a field containing a root of x^3+x+1 over $\mathbb{F}_2$ and list a basis over $\mathbb{F}_2$.
Hint 1 — a starting point. First check irreducibility using the two possible roots.
Hint 2 — the key move. Use the quotient by the polynomial.

MM5020-U7.10 · A · 50 min
Learning objective: U7.1 Compositum degree
Prove $\left[\mathbb{Q}\left(\sqrt{2},\sqrt{5}\right) : \mathbb{Q}\right] = 4$ and find a basis.
Hint 1 — a starting point. Show $\sqrt{5}$ is not in $\mathbb{Q}\left(\sqrt{2}\right)$.
Hint 2 — the key move. Square a proposed expression $a+b\sqrt{2}$.

MM5020-U7.11 · A · 50 min
Learning objective: U7.2 Transitivity of algebraicity
Let $\frac{L}{K}$ be algebraic and let α be algebraic over L. Prove α is algebraic over K, even when $\frac{L}{K}$ is infinite.
Hint 1 — a starting point. Only finitely many coefficients occur in an annihilating polynomial.
Hint 2 — the key move. Adjoin those coefficients to K first.

MM5020-U7.12 · A · 50 min
Learning objective: U7.2 A primitive generator
Show $\mathbb{Q}\left(\sqrt{2}+\sqrt{3}\right) = \mathbb{Q}\left(\sqrt{2},\sqrt{3}\right)$ and find the minimal polynomial of the sum.
Hint 1 — a starting point. The reciprocal of the sum recovers the difference.
Hint 2 — the key move. Since $\left(\sqrt{3}-\sqrt{2}\right)\left(\sqrt{3}+\sqrt{2}\right) = 1$, both roots can be recovered.

8 Unit 8 · Finite fields and applications

MM5020-U8.01 · E · 15 min
Learning objective: U8.1 Field orders
Which of the orders $8, 9, 10, 16, 18, 25$ can occur for a finite field?
Hint 1 — a starting point. Finite-field orders are prime powers.
Hint 2 — the key move. Apply the existence theorem as well as the necessary condition.

MM5020-U8.02 · E · 15 min
Learning objective: U8.1 Arithmetic in four elements
In $\mathbb{F}_4 = \mathbb{F}_2[\alpha]$ with $\alpha^2 + \alpha + 1 = 0$, compute $\alpha(\alpha + 1)$ and the inverse of $\alpha + 1$.
Hint 1 — a starting point. Use $\alpha^2 = \alpha + 1$.
Hint 2 — the key move. Addition has characteristic two.

MM5020-U8.03 · E · 15 min
Learning objective: U8.2 Generators
Find the order of two in $\mathbb{F}_7^{\times}$.
Hint 1 — a starting point. Compute successive powers.
Hint 2 — the key move. The third power is one.

MM5020-U8.04 · E · 15 min
Learning objective: U8.1 Characteristic and dimension
What are the characteristic and dimension over the prime field of a field of order eighty-one?
Hint 1 — a starting point. Write the order as a prime power.
Hint 2 — the key move. Eighty-one equals three to the fourth power.

MM5020-U8.05 · C · 30 min
Learning objective: U8.2 Frobenius
Prove the map $a \mapsto a^p$ is an automorphism of a finite field of characteristic p.
Hint 1 — a starting point. Check addition, multiplication, and injectivity.
Hint 2 — the key move. Use the binomial theorem and the fact a field has no nonzero nilpotents.

MM5020-U8.06 · C · 30 min
Learning objective: U8.2 Interpolation
Find the degree-less-than-three polynomial over $\mathbb{F}_5$ with $f(0) = 1$, $f(1) = 0$, $f(2) = 2$.
Hint 1 — a starting point. Write $f = ax^2 + bx + c$.
Hint 2 — the key move. The first value gives $c = 1$; solve the remaining two equations modulo five.

MM5020-U8.07 · C · 30 min
Learning objective: U8.1 Polynomial roots
Show every element of $\mathbb{F}_q$ is a root of $x^q - x$, and this polynomial has no repeated root.
Hint 1 — a starting point. Apply Lagrange to the nonzero elements.
Hint 2 — the key move. Its formal derivative is negative one.

MM5020-U8.08 · C · 30 min
Learning objective: U8.2 Evaluation codes
Evaluate all degree-less-than-two polynomials at the five elements of $\mathbb{F}_5$. How many codewords result, and what is their minimum distance?
Hint 1 — a starting point. There are two free field coefficients.
Hint 2 — the key move. A nonzero difference is a nonzero polynomial of degree at most one.

MM5020-U8.09 · C · 30 min
Learning objective: U8.2 Primitive elements
Find all generators of $\mathbb{F}_7^\times$, given that three is one generator.
Hint 1 — a starting point. In a cyclic group of order six, generator powers have exponents coprime to six.
Hint 2 — the key move. Use exponents one and five.

MM5020-U8.10 · A · 50 min
Learning objective: U8.1 Subfields
Show a field of sixteen elements has a unique subfield of four elements.
Hint 1 — a starting point. Use the roots of $x^4 - x$ inside the larger field.
Hint 2 — the key move. Its nonzero roots satisfy $x^3 = 1$ in a cyclic group of order fifteen.

MM5020-U8.11 · A · 50 min
Learning objective: U8.2 Polynomial functions
Prove every function $\mathbb{F}_q \to \mathbb{F}_q$ is represented by a unique polynomial of degree less than q. Is the representing polynomial unique without the degree bound?
Hint 1 — a starting point. Interpolate at every element of the field.
Hint 2 — the key move. The polynomial $x^q - x$ represents the zero function.

MM5020-U8.12 · A · 50 min
Learning objective: U8.1 Irreducible quadratics
Count the monic irreducible quadratics over $\mathbb{F}_q$.
Hint 1 — a starting point. Count all monic quadratics and subtract the reducible ones.
Hint 2 — the key move. Reducible monic quadratics correspond to unordered pairs of roots, allowing repetition.

9 Mock assessment 1

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM5020-M1.01 · E · 15 min
Learning objective: U1–U2 Cyclic groups
List the orders of every element of $\mathbb{Z}/8\mathbb{Z}$.
Hint 1 — a starting point. Use the gcd formula.
Hint 2 — the key move. For residue k, the order is $\frac{8}{\gcd(8,k)}$.

MM5020-M1.02 · E · 15 min
Learning objective: U5 Ring quotients
Determine whether $\mathbb{Z}/15\mathbb{Z}$ is a field and list its units.
Hint 1 — a starting point. Check coprimality to fifteen.
Hint 2 — the key move. The product of three and five is zero.

MM5020-M1.03 · C · 30 min
Learning objective: U3–U4 Group structure
Let $|G| = 49$. Prove G is abelian, and show that if some element has order forty-nine then G is cyclic.
Hint 1 — a starting point. Use the center of a prime-power group.
Hint 2 — the key move. The center has order seven or forty-nine.

MM5020-M1.04 · C · 30 min
Learning objective: U6–U7 Algebraic extensions
Let $\alpha^3 = 3$ over $\mathbb{Q}$. Prove the extension degree is three and find $(\alpha - 1)^{-1}$.
Hint 1 — a starting point. Use Eisenstein and a geometric-product identity.
Hint 2 — the key move. The product with $\alpha^2 + \alpha + 1$ is two.

MM5020-M1.05 · A · 50 min
Learning objective: U3 Universal constructions
For normal subgroups N, M of G, show $g \mapsto (gN, gM)$ induces an injective homomorphism $G/(N \cap M) \to G/N \times G/M$. When is it surjective?
Hint 1 — a starting point. Compute the kernel of the paired map.
Hint 2 — the key move. Surjectivity is equivalent to $G = NM$.

MM5020-M1.06 · A · 50 min
Learning objective: U8 Field structure
Prove that the fixed field of Frobenius $x \mapsto x^p$ in $\mathbb{F}_{p^n}$ is exactly $\mathbb{F}_p$.
Hint 1 — a starting point. Fixed elements are roots of $x^p - x$.
Hint 2 — the key move. The prime field already supplies p roots.

10 Mock assessment 2

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM5020-M2.01 · E · 15 min
Learning objective: U2 Permutations
Find the order and sign of $(123)(45)(67)$.
Hint 1 — a starting point. Use disjoint cycles.
Hint 2 — the key move. Take the least common multiple of three, two, and two.

MM5020-M2.02 · E · 15 min
Learning objective: U7 Minimal polynomials
Find the minimal polynomial of $2+\sqrt{5}$ over $\mathbb{Q}$.
Hint 1 — a starting point. Isolate the square root.
Hint 2 — the key move. Use $(x-2)^2-5$.

MM5020-M2.03 · C · 30 min
Learning objective: U3 Quotient isomorphisms
Prove $S_4/A_4 \cong C_2$.
Hint 1 — a starting point. Use sign.
Hint 2 — the key move. Its kernel is A_4 and a transposition maps to negative one.

MM5020-M2.04 · C · 30 min
Learning objective: U5–U6 Field construction
Prove $\mathbb{F}_3[x]/(x^2+1)$ is a field and find the inverse of $[x+1]$.
Hint 1 — a starting point. Check roots, then multiply by $x-1$.
Hint 2 — the key move. Modulo x^2+1, the product is negative two, which equals one in $\mathbb{F}_3$.

MM5020-M2.05 · A · 50 min
Learning objective: U4 Actions
A group G of order fifteen acts on a set of seven elements. Prove there is a fixed point.
Hint 1 — a starting point. Orbit sizes divide fifteen.
Hint 2 — the key move. Only sizes one, three, and five fit inside seven.

MM5020-M2.06 · A · 50 min
Learning objective: U7–U8 Degrees and subfields
Prove that $\mathbb{F}_{p^3}$ has no subfield strictly between $\mathbb{F}_p$ and itself.
Hint 1 — a starting point. Translate subfield cardinalities into vector-space degrees.
Hint 2 — the key move. The tower law makes the intermediate degree divide three.